ActionShap: Supplementary Material

Sections S1–S10

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This reviewer-visible supplementary material should be read alongside the main manuscript. It provides the expanded numerical audit underlying the compact manuscript tables, including robustness matrices, statistical validation, replication analyses, and computational-cost details. The tables are generated from the schema-v2 result archive. The accompanying artifact will record the source revision, archive hash, dependency manifest, and machine-readable CSV/JSON results.

CCS Concepts: • **Information systems** → **Recommender systems**; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: recommender systems, explanation evaluation, bounded intervention, robustness, reproducibility

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S1 Alignment matrix (Table S1)

Table S1. Complete singleton alignment matrix for all declared ItemKNN conditions. Means are distinct-user values; confidence intervals and missing-user counts are retained in the CSV release.

Condition	Method	Deletion	Bounded	Gap	n
Amazon full catalogue	Greedy sequential deletion	0.5731	0.4494	-0.1236	250
Amazon full catalogue	LIME	0.9215	0.7215	-0.2000	250
Amazon full catalogue	Leave-one-out	1.0000	0.7895	-0.2105	250
Amazon full catalogue	Monte Carlo Shapley	-0.2875	-0.1186	0.1689	250
Amazon full catalogue	Random control	-0.0012	-0.0041	-0.0028	250
Amazon sampled	Greedy sequential deletion	0.1911	0.1310	-0.0600	993
Amazon sampled	LIME	0.9304	0.8271	-0.1032	993

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Condition	Method	Deletion	Bounded	Gap	n
Amazon sampled	Leave-one-out	1.0000	0.8511	-0.1489	993
Amazon sampled	Monte Carlo Shapley	0.2856	0.4141	0.1285	993
Amazon sampled	Random control	-0.0080	-0.0103	-0.0023	993
MovieLens 100 candidates	Greedy sequential deletion	0.2632	0.2605	-0.0027	250
MovieLens 100 candidates	LIME	0.9604	0.9469	-0.0136	250
MovieLens 100 candidates	Leave-one-out	1.0000	0.9824	-0.0176	250
MovieLens 100 candidates	Monte Carlo Shapley	0.8086	0.8209	0.0124	250
MovieLens 100 candidates	Random control	-0.0044	-0.0029	0.0015	250
MovieLens 500 candidates	Greedy sequential deletion	0.3733	0.3782	0.0049	250
MovieLens 500 candidates	LIME	0.9445	0.9219	-0.0226	250
MovieLens 500 candidates	Leave-one-out	1.0000	0.9710	-0.0290	250
MovieLens 500 candidates	Monte Carlo Shapley	0.6918	0.7239	0.0321	250
MovieLens 500 candidates	Random control	0.0018	0.0040	0.0022	250
MovieLens full catalogue	Greedy sequential deletion	0.3845	0.3819	-0.0026	250
MovieLens full catalogue	LIME	0.9256	0.8924	-0.0332	250
MovieLens full catalogue	Leave-one-out	1.0000	0.9580	-0.0420	250
MovieLens full catalogue	Monte Carlo Shapley	0.5742	0.6180	0.0438	250
MovieLens full catalogue	Random control	0.0033	0.0039	0.0006	250
MovieLens n_max=100	Greedy sequential deletion	0.3874	0.3807	-0.0067	250
MovieLens n_max=100	LIME	0.9280	0.9183	-0.0097	250
MovieLens n_max=100	Leave-one-out	1.0000	0.9872	-0.0128	250
MovieLens n_max=100	Monte Carlo Shapley	0.8018	0.8082	0.0064	250
MovieLens n_max=100	Random control	0.0010	0.0015	0.0005	250
MovieLens n_max=50	Greedy sequential deletion	0.3694	0.3636	-0.0058	250
MovieLens n_max=50	LIME	0.9448	0.9344	-0.0105	250
MovieLens n_max=50	Leave-one-out	1.0000	0.9867	-0.0133	250

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Condition	Method	Deletion	Bounded	Gap	n
MovieLens n_max=50	Monte Carlo Shapley	0.7997	0.8079	0.0082	250
MovieLens n_max=50	Random control	0.0010	0.0013	0.0003	250
MovieLens rho=0.25	Greedy sequential deletion	0.3780	0.3766	-0.0014	250
MovieLens rho=0.25	LIME	0.9551	0.9498	-0.0053	250
MovieLens rho=0.25	Leave-one-out	1.0000	0.9922	-0.0078	250
MovieLens rho=0.25	Monte Carlo Shapley	0.7596	0.7664	0.0068	250
MovieLens rho=0.25	Random control	-0.0038	-0.0050	-0.0012	250
MovieLens sampled	Greedy sequential deletion	0.3205	0.3183	-0.0023	1000
MovieLens sampled	LIME	0.9509	0.9334	-0.0176	1000
MovieLens sampled	Leave-one-out	1.0000	0.9780	-0.0220	1000
MovieLens sampled	Monte Carlo Shapley	0.7617	0.7786	0.0169	1000
MovieLens sampled	Random control	-0.0010	0.0003	0.0014	1000

S2 Joint outcomes (Table S2)

Table S2. Complete joint-intervention outcome matrix. The metric name identifies the utility and whether the quantity is an effect, success, abstention, or regret.

Condition	Dataset	Method	Metric	Mean	*n*
budget1	MovieLens	Greedy sequential deletion	abstention	0.0000	250
budget1	MovieLens	Greedy sequential deletion	intervention success ndcg	0.0640	250
budget1	MovieLens	Greedy sequential deletion	joint effect ndcg	-0.0016	250
budget1	MovieLens	Greedy sequential deletion	normalized regret ndcg	0.9437	60
budget1	MovieLens	LIME	abstention	0.0040	250
budget1	MovieLens	LIME	intervention success ndcg	0.1816	250
budget1	MovieLens	LIME	joint effect ndcg	0.0260	250
budget1	MovieLens	LIME	normalized regret ndcg	0.2966	60
budget1	MovieLens	Leave-one-out	abstention	0.0040	250
budget1	MovieLens	Leave-one-out	intervention success ndcg	0.1880	250
budget1	MovieLens	Leave-one-out	joint effect ndcg	0.0268	250
budget1	MovieLens	Leave-one-out	normalized regret ndcg	0.2601	60
budget1	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
budget1	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.1944	250
budget1	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0264	250
budget1	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2738	60
budget1	MovieLens	Random control	abstention	0.0008	250
budget1	MovieLens	Random control	intervention success ndcg	0.0456	250
budget1	MovieLens	Random control	joint effect ndcg	0.0009	250
budget1	MovieLens	Random control	normalized regret ndcg	0.9598	60
budget3	MovieLens	Greedy sequential deletion	abstention	0.0000	250

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Condition	Dataset	Method	Metric	Mean	n
budget3	MovieLens	Greedy sequential deletion	intervention success ndcg	0.1520	250
budget3	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0097	250
budget3	MovieLens	LIME	abstention	0.0040	250
budget3	MovieLens	LIME	intervention success ndcg	0.3400	250
budget3	MovieLens	LIME	joint effect ndcg	0.0620	250
budget3	MovieLens	Leave-one-out	abstention	0.0040	250
budget3	MovieLens	Leave-one-out	intervention success ndcg	0.3400	250
budget3	MovieLens	Leave-one-out	joint effect ndcg	0.0615	250
budget3	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
budget3	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.3400	250
budget3	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0622	250
budget3	MovieLens	Random control	abstention	0.0008	250
budget3	MovieLens	Random control	intervention success ndcg	0.0952	250
budget3	MovieLens	Random control	joint effect ndcg	0.0019	250
candidates100	MovieLens	Greedy sequential deletion	abstention	0.0000	250
candidates100	MovieLens	Greedy sequential deletion	intervention success ndcg	0.0800	250
candidates100	MovieLens	Greedy sequential deletion	joint effect ndcg	-0.0030	250
candidates100	MovieLens	Greedy sequential deletion	normalized regret ndcg	1.1995	94
candidates100	MovieLens	LIME	abstention	0.0000	250
candidates100	MovieLens	LIME	intervention success ndcg	0.3344	250
candidates100	MovieLens	LIME	joint effect ndcg	0.0546	250
candidates100	MovieLens	LIME	normalized regret ndcg	0.1566	94
candidates100	MovieLens	Leave-one-out	abstention	0.0000	250
candidates100	MovieLens	Leave-one-out	intervention success ndcg	0.3400	250

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Condition	Dataset	Method	Metric	Mean	*n*
candidates100	MovieLens	Leave-one-out	joint effect ndcg	0.0539	250
candidates100	MovieLens	Leave-one-out	normalized regret ndcg	0.1373	94
candidates100	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
candidates100	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.3320	250
candidates100	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0547	250
candidates100	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.1654	94
candidates100	MovieLens	Random control	abstention	0.0008	250
candidates100	MovieLens	Random control	intervention success ndcg	0.0960	250
candidates100	MovieLens	Random control	joint effect ndcg	0.0057	250
candidates100	MovieLens	Random control	normalized regret ndcg	1.1430	94
candidates500	MovieLens	Greedy sequential deletion	abstention	0.0000	250
candidates500	MovieLens	Greedy sequential deletion	intervention success ndcg	0.1000	250
candidates500	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0065	250
candidates500	MovieLens	Greedy sequential deletion	normalized regret ndcg	0.9937	60
candidates500	MovieLens	LIME	abstention	0.0000	250
candidates500	MovieLens	LIME	intervention success ndcg	0.2120	250
candidates500	MovieLens	LIME	joint effect ndcg	0.0297	250
candidates500	MovieLens	LIME	normalized regret ndcg	0.1702	60
candidates500	MovieLens	Leave-one-out	abstention	0.0000	250
candidates500	MovieLens	Leave-one-out	intervention success ndcg	0.2120	250
candidates500	MovieLens	Leave-one-out	joint effect ndcg	0.0287	250
candidates500	MovieLens	Leave-one-out	normalized regret ndcg	0.1821	60
candidates500	MovieLens	Monte Carlo Shapley	abstention	0.0000	250

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Condition	Dataset	Method	Metric	Mean	*n*
candidates500	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.1952	250
candidates500	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0250	250
candidates500	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2840	60
candidates500	MovieLens	Random control	abstention	0.0008	250
candidates500	MovieLens	Random control	intervention success ndcg	0.0568	250
candidates500	MovieLens	Random control	joint effect ndcg	0.0012	250
candidates500	MovieLens	Random control	normalized regret ndcg	1.4627	60
full_catalogue	Amazon	Greedy sequential deletion	abstention	0.0000	250
full_catalogue	Amazon	Greedy sequential deletion	intervention success ndcg	0.0120	250
full_catalogue	Amazon	Greedy sequential deletion	joint effect ndcg	0.0025	250
full_catalogue	Amazon	Greedy sequential deletion	normalized regret ndcg	0.4194	5
full_catalogue	Amazon	LIME	abstention	0.4608	250
full_catalogue	Amazon	LIME	intervention success ndcg	0.0120	250
full_catalogue	Amazon	LIME	joint effect ndcg	0.0026	250
full_catalogue	Amazon	LIME	normalized regret ndcg	0.4000	5
full_catalogue	Amazon	Leave-one-out	abstention	0.4560	250
full_catalogue	Amazon	Leave-one-out	intervention success ndcg	0.0120	250
full_catalogue	Amazon	Leave-one-out	joint effect ndcg	0.0026	250
full_catalogue	Amazon	Leave-one-out	normalized regret ndcg	0.4000	5
full_catalogue	Amazon	Monte Carlo Shapley	abstention	0.0000	250
full_catalogue	Amazon	Monte Carlo Shapley	intervention success ndcg	0.0152	250
full_catalogue	Amazon	Monte Carlo Shapley	joint effect ndcg	0.0035	250

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Condition	Dataset	Method	Metric	Mean	*n*
full_catalogue	Amazon	Monte Carlo Shapley	normalized regret ndcg	0.2555	5
full_catalogue	Amazon	Random control	abstention	0.0528	250
full_catalogue	Amazon	Random control	intervention success ndcg	0.0048	250
full_catalogue	Amazon	Random control	joint effect ndcg	0.0005	250
full_catalogue	Amazon	Random control	normalized regret ndcg	0.7928	5
full_catalogue	MovieLens	Greedy sequential deletion	abstention	0.0000	250
full_catalogue	MovieLens	Greedy sequential deletion	intervention success ndcg	0.0440	250
full_catalogue	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0046	250
full_catalogue	MovieLens	Greedy sequential deletion	normalized regret ndcg	0.6189	20
full_catalogue	MovieLens	LIME	abstention	0.0000	250
full_catalogue	MovieLens	LIME	intervention success ndcg	0.0720	250
full_catalogue	MovieLens	LIME	joint effect ndcg	0.0145	250
full_catalogue	MovieLens	LIME	normalized regret ndcg	0.2076	20
full_catalogue	MovieLens	Leave-one-out	abstention	0.0000	250
full_catalogue	MovieLens	Leave-one-out	intervention success ndcg	0.0720	250
full_catalogue	MovieLens	Leave-one-out	joint effect ndcg	0.0143	250
full_catalogue	MovieLens	Leave-one-out	normalized regret ndcg	0.2133	20
full_catalogue	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
full_catalogue	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.0672	250
full_catalogue	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0115	250
full_catalogue	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2905	20
full_catalogue	MovieLens	Random control	abstention	0.0008	250
full_catalogue	MovieLens	Random control	intervention success ndcg	0.0208	250
full_catalogue	MovieLens	Random control	joint effect ndcg	-0.0005	250

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Condition	Dataset	Method	Metric	Mean	*n*
full_catalogue	MovieLens	Random control	normalized regret ndcg	1.2435	20
nmax100	MovieLens	Greedy sequential deletion	abstention	0.0000	250
nmax100	MovieLens	Greedy sequential deletion	intervention success ndcg	0.0640	250
nmax100	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0028	250
nmax100	MovieLens	Greedy sequential deletion	normalized regret ndcg	0.8313	55
nmax100	MovieLens	LIME	abstention	0.0040	250
nmax100	MovieLens	LIME	intervention success ndcg	0.1720	250
nmax100	MovieLens	LIME	joint effect ndcg	0.0221	250
nmax100	MovieLens	LIME	normalized regret ndcg	0.2604	55
nmax100	MovieLens	Leave-one-out	abstention	0.0040	250
nmax100	MovieLens	Leave-one-out	intervention success ndcg	0.1760	250
nmax100	MovieLens	Leave-one-out	joint effect ndcg	0.0219	250
nmax100	MovieLens	Leave-one-out	normalized regret ndcg	0.2449	55
nmax100	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
nmax100	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.1784	250
nmax100	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0237	250
nmax100	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2283	55
nmax100	MovieLens	Random control	abstention	0.0008	250
nmax100	MovieLens	Random control	intervention success ndcg	0.0376	250
nmax100	MovieLens	Random control	joint effect ndcg	-0.0025	250
nmax100	MovieLens	Random control	normalized regret ndcg	1.1975	55
nmax50	MovieLens	Greedy sequential deletion	abstention	0.0000	250
nmax50	MovieLens	Greedy sequential deletion	intervention success ndcg	0.0680	250

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Condition	Dataset	Method	Metric	Mean	*n*
nmax50	MovieLens	Greedy sequential deletion	joint effect ndcg	-0.0050	250
nmax50	MovieLens	Greedy sequential deletion	normalized regret ndcg	2.0846	61
nmax50	MovieLens	LIME	abstention	0.0040	250
nmax50	MovieLens	LIME	intervention success ndcg	0.2096	250
nmax50	MovieLens	LIME	joint effect ndcg	0.0298	250
nmax50	MovieLens	LIME	normalized regret ndcg	0.2090	61
nmax50	MovieLens	Leave-one-out	abstention	0.0040	250
nmax50	MovieLens	Leave-one-out	intervention success ndcg	0.2120	250
nmax50	MovieLens	Leave-one-out	joint effect ndcg	0.0295	250
nmax50	MovieLens	Leave-one-out	normalized regret ndcg	0.2150	61
nmax50	MovieLens	Monte Carlo Shapley	abstention	0.0000	250
nmax50	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.2048	250
nmax50	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0295	250
nmax50	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2271	61
nmax50	MovieLens	Random control	abstention	0.0008	250
nmax50	MovieLens	Random control	intervention success ndcg	0.0512	250
nmax50	MovieLens	Random control	joint effect ndcg	-0.0024	250
nmax50	MovieLens	Random control	normalized regret ndcg	1.5288	61
primary	Amazon	Greedy sequential deletion	abstention	0.0070	1000
primary	Amazon	Greedy sequential deletion	intervention success ndcg	0.0900	1000
primary	Amazon	Greedy sequential deletion	joint effect ndcg	0.0102	1000
primary	Amazon	Greedy sequential deletion	normalized regret ndcg	0.7836	196
primary	Amazon	LIME	abstention	0.0702	1000

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Table S2 – continued from previous page

Condition	Dataset	Method	Metric	Mean	n
primary	Amazon	LIME	intervention success ndcg	0.1712	1000
primary	Amazon	LIME	joint effect ndcg	0.0331	1000
primary	Amazon	LIME	normalized regret ndcg	0.1975	196
primary	Amazon	Leave-one-out	abstention	0.0860	1000
primary	Amazon	Leave-one-out	intervention success ndcg	0.1670	1000
primary	Amazon	Leave-one-out	joint effect ndcg	0.0313	1000
primary	Amazon	Leave-one-out	normalized regret ndcg	0.2325	196
primary	Amazon	Monte Carlo Shapley	abstention	0.0102	1000
primary	Amazon	Monte Carlo Shapley	intervention success ndcg	0.1702	1000
primary	Amazon	Monte Carlo Shapley	joint effect ndcg	0.0333	1000
primary	Amazon	Monte Carlo Shapley	normalized regret ndcg	0.2061	196
primary	Amazon	Random control	abstention	0.0476	1000
primary	Amazon	Random control	intervention success ndcg	0.0660	1000
primary	Amazon	Random control	joint effect ndcg	-0.0039	1000
primary	Amazon	Random control	normalized regret ndcg	1.1971	196
primary	MovieLens	Greedy sequential deletion	abstention	0.0000	1000
primary	MovieLens	Greedy sequential deletion	intervention success ndcg	0.1030	1000
primary	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0058	1000
primary	MovieLens	Greedy sequential deletion	normalized regret ndcg	1.4710	339
primary	MovieLens	LIME	abstention	0.0000	1000
primary	MovieLens	LIME	intervention success ndcg	0.2876	1000
primary	MovieLens	LIME	joint effect ndcg	0.0425	1000
primary	MovieLens	LIME	normalized regret ndcg	0.2254	339
primary	MovieLens	Leave-one-out	abstention	0.0000	1000
primary	MovieLens	Leave-one-out	intervention success ndcg	0.2890	1000

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Condition	Dataset	Method	Metric	Mean	*n*
primary	MovieLens	Leave-one-out	joint effect ndcg	0.0431	1000
primary	MovieLens	Leave-one-out	normalized regret ndcg	0.2174	339
primary	MovieLens	Monte Carlo Shapley	abstention	0.0000	1000
primary	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.2742	1000
primary	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0403	1000
primary	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.2675	339
primary	MovieLens	Random control	abstention	0.0004	1000
primary	MovieLens	Random control	intervention success ndcg	0.0744	1000
primary	MovieLens	Random control	joint effect ndcg	-0.0010	1000
primary	MovieLens	Random control	normalized regret ndcg	1.4455	339
rho025	MovieLens	Greedy sequential deletion	abstention	0.0000	250
rho025	MovieLens	Greedy sequential deletion	intervention success ndcg	0.1440	250
rho025	MovieLens	Greedy sequential deletion	joint effect ndcg	0.0060	250
rho025	MovieLens	Greedy sequential deletion	normalized regret ndcg	0.9987	98
rho025	MovieLens	LIME	abstention	0.0040	250
rho025	MovieLens	LIME	intervention success ndcg	0.3576	250
rho025	MovieLens	LIME	joint effect ndcg	0.0651	250
rho025	MovieLens	LIME	normalized regret ndcg	0.1565	98
rho025	MovieLens	Leave-one-out	abstention	0.0040	250
rho025	MovieLens	Leave-one-out	intervention success ndcg	0.3600	250
rho025	MovieLens	Leave-one-out	joint effect ndcg	0.0653	250
rho025	MovieLens	Leave-one-out	normalized regret ndcg	0.1440	98
rho025	MovieLens	Monte Carlo Shapley	abstention	0.0000	250

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Condition	Dataset	Method	Metric	Mean	*n*
rho025	MovieLens	Monte Carlo Shapley	intervention success ndcg	0.3584	250
rho025	MovieLens	Monte Carlo Shapley	joint effect ndcg	0.0643	250
rho025	MovieLens	Monte Carlo Shapley	normalized regret ndcg	0.1691	98
rho025	MovieLens	Random control	abstention	0.0008	250
rho025	MovieLens	Random control	intervention success ndcg	0.0944	250
rho025	MovieLens	Random control	joint effect ndcg	0.0002	250
rho025	MovieLens	Random control	normalized regret ndcg	1.2587	98

S3 Paired comparisons (Tables S3–S5)

Table S3: AIA and gap comparisons

Table S3. deletion AIA, bounded AIA, and Actionability Gap comparisons.

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
Amazon	Actionability Gap	Greedy seq. del.	Random	-0.0577	-0.0750	-0.0411	0.0010	-0.2104	993
Amazon	Actionability Gap	LIME	Greedy seq. del.	-0.0432	-0.0586	-0.0273	0.0010	-0.1695	993
Amazon	Actionability Gap	LIME	LOO	0.0457	0.0365	0.0555	0.0010	0.3006	993
Amazon	Actionability Gap	LIME	Random	-0.1009	-0.1163	-0.0860	0.0010	-0.4122	993
Amazon	Actionability Gap	LOO	Greedy seq. del.	-0.0888	-0.1054	-0.0730	0.0010	-0.3373	993
Amazon	Actionability Gap	LOO	Random	-0.1465	-0.1633	-0.1304	0.0010	-0.5540	993
Amazon	Actionability Gap	MC Shapley	Greedy seq. del.	0.1886	0.1594	0.2198	0.0010	0.3859	993
Amazon	Actionability Gap	MC Shapley	LIME	0.2318	0.2021	0.2629	0.0010	0.4729	993
Amazon	Actionability Gap	MC Shapley	LOO	0.2774	0.2452	0.3109	0.0010	0.5248	993
Amazon	Actionability Gap	MC Shapley	Random	0.1309	0.1108	0.1515	0.0010	0.3920	993
Amazon	bounded AIA	Greedy seq. del.	Random	0.1413	0.1058	0.1766	0.0010	0.2492	993
Amazon	bounded AIA	LIME	Greedy seq. del.	0.6961	0.6638	0.7301	0.0010	1.3169	993
Amazon	bounded AIA	LIME	LOO	-0.0240	-0.0317	-0.0163	0.0010	-0.1908	993
Amazon	bounded AIA	LIME	Random	0.8375	0.8166	0.8589	0.0010	2.4300	993
Amazon	bounded AIA	LOO	Greedy seq. del.	0.7201	0.6878	0.7531	0.0010	1.3796	993
Amazon	bounded AIA	LOO	Random	0.8614	0.8394	0.8830	0.0010	2.4605	993
Amazon	bounded AIA	MC Shapley	Greedy seq. del.	0.2831	0.2377	0.3278	0.0010	0.3878	993
Amazon	bounded AIA	MC Shapley	LIME	-0.4130	-0.4489	-0.3782	0.0010	-0.7258	993
Amazon	bounded AIA	MC Shapley	LOO	-0.4370	-0.4721	-0.4019	0.0010	-0.7711	993
Amazon	bounded AIA	MC Shapley	Random	0.4244	0.3874	0.4608	0.0010	0.7255	993
Amazon	deletion AIA	Greedy seq. del.	Random	0.1990	0.1642	0.2342	0.0010	0.3561	993
Amazon	deletion AIA	LIME	Greedy seq. del.	0.7393	0.7080	0.7710	0.0010	1.4431	993
Amazon	deletion AIA	LIME	LOO	-0.0696	-0.0762	-0.0635	0.0010	-0.6841	993
Amazon	deletion AIA	LIME	Random	0.9383	0.9221	0.9546	0.0010	3.6155	993
Amazon	deletion AIA	LOO	Greedy seq. del.	0.8089	0.7767	0.8406	0.0010	1.5799	993
Amazon	deletion AIA	LOO	Random	1.0080	0.9934	1.0226	0.0010	4.2381	993
Amazon	deletion AIA	MC Shapley	Greedy seq. del.	0.0945	0.0451	0.1429	0.0010	0.1211	993
Amazon	deletion AIA	MC Shapley	LIME	-0.6448	-0.6836	-0.6083	0.0010	-1.0634	993
Amazon	deletion AIA	MC Shapley	LOO	-0.7144	-0.7523	-0.6766	0.0010	-1.1771	993
Amazon	deletion AIA	MC Shapley	Random	0.2935	0.2521	0.3340	0.0010	0.4488	993

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Table S3 – continued from previous page

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
MovieLens	Actionability Gap	Greedy seq. del.	Random	-0.0036	-0.0109	0.0018	0.3099	-0.0351	1000
MovieLens	Actionability Gap	LIME	Greedy seq. del.	-0.0153	-0.0194	-0.0115	0.0010	-0.2410	1000
MovieLens	Actionability Gap	LIME	LOO	0.0045	0.0031	0.0060	0.0010	0.1942	1000
MovieLens	Actionability Gap	LIME	Random	-0.0190	-0.0260	-0.0142	0.0010	-0.1958	1000
MovieLens	Actionability Gap	LOO	Greedy seq. del.	-0.0198	-0.0237	-0.0160	0.0010	-0.3175	1000
MovieLens	Actionability Gap	LOO	Random	-0.0234	-0.0303	-0.0186	0.0010	-0.2427	1000
MovieLens	Actionability Gap	MC Shapley	Greedy seq. del.	0.0191	0.0127	0.0273	0.0010	0.1596	1000
MovieLens	Actionability Gap	MC Shapley	LIME	0.0344	0.0281	0.0428	0.0010	0.2888	1000
MovieLens	Actionability Gap	MC Shapley	LOO	0.0389	0.0323	0.0475	0.0010	0.3194	1000
MovieLens	Actionability Gap	MC Shapley	Random	0.0155	0.0119	0.0193	0.0010	0.2611	1000
MovieLens	bounded AIA	Greedy seq. del.	Random	0.3179	0.3000	0.3357	0.0010	1.0816	1000
MovieLens	bounded AIA	LIME	Greedy seq. del.	0.6151	0.5976	0.6327	0.0010	2.1833	1000
MovieLens	bounded AIA	LIME	LOO	-0.0446	-0.0476	-0.0416	0.0010	-0.9264	1000
MovieLens	bounded AIA	LIME	Random	0.9330	0.9241	0.9414	0.0010	6.5305	1000
MovieLens	bounded AIA	LOO	Greedy seq. del.	0.6597	0.6425	0.6769	0.0010	2.3881	1000
MovieLens	bounded AIA	LOO	Random	0.9776	0.9696	0.9856	0.0010	7.6132	1000
MovieLens	bounded AIA	MC Shapley	Greedy seq. del.	0.4604	0.4402	0.4801	0.0010	1.4312	1000
MovieLens	bounded AIA	MC Shapley	LIME	-0.1547	-0.1643	-0.1452	0.0010	-1.0100	1000
MovieLens	bounded AIA	MC Shapley	LOO	-0.1994	-0.2095	-0.1896	0.0010	-1.2542	1000
MovieLens	bounded AIA	MC Shapley	Random	0.7783	0.7662	0.7897	0.0010	4.0791	1000
MovieLens	deletion AIA	Greedy seq. del.	Random	0.3216	0.3031	0.3405	0.0010	1.0794	1000
MovieLens	deletion AIA	LIME	Greedy seq. del.	0.6304	0.6136	0.6475	0.0010	2.2535	1000
MovieLens	deletion AIA	LIME	LOO	-0.0491	-0.0518	-0.0464	0.0010	-1.1099	1000
MovieLens	deletion AIA	LIME	Random	0.9520	0.9446	0.9593	0.0010	8.0043	1000
MovieLens	deletion AIA	LOO	Greedy seq. del.	0.6795	0.6623	0.6964	0.0010	2.4705	1000
MovieLens	deletion AIA	LOO	Random	1.0010	0.9943	1.0080	0.0010	8.9894	1000
MovieLens	deletion AIA	MC Shapley	Greedy seq. del.	0.4412	0.4200	0.4619	0.0010	1.2995	1000
MovieLens	deletion AIA	MC Shapley	LIME	-0.1892	-0.2001	-0.1787	0.0010	-1.1174	1000
MovieLens	deletion AIA	MC Shapley	LOO	-0.2383	-0.2494	-0.2271	0.0010	-1.3229	1000
MovieLens	deletion AIA	MC Shapley	Random	0.7628	0.7499	0.7753	0.0010	3.7061	1000

Table S4: joint NDCG effect and success comparisons

Table S4. joint NDCG effect and success comparisons.

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
Amazon	Success	Greedy seq. del.	Random	0.0240	0.0106	0.0376	0.0036	0.1094	1000
Amazon	Success	LIME	Greedy seq. del.	0.0812	0.0628	0.1000	0.0010	0.2684	1000
Amazon	Success	LIME	LOO	0.0042	0.0002	0.0092	0.2256	0.0572	1000
Amazon	Success	LIME	Random	0.1052	0.0888	0.1224	0.0010	0.3941	1000
Amazon	Success	LOO	Greedy seq. del.	0.0770	0.0590	0.0960	0.0010	0.2550	1000
Amazon	Success	LOO	Random	0.1010	0.0848	0.1180	0.0010	0.3766	1000
Amazon	Success	MC Shapley	Greedy seq. del.	0.0802	0.0622	0.0990	0.0010	0.2684	1000
Amazon	Success	MC Shapley	LIME	-0.0010	-0.0072	0.0054	0.9045	-0.0099	1000
Amazon	Success	MC Shapley	LOO	0.0032	-0.0046	0.0112	0.9045	0.0251	1000
Amazon	Success	MC Shapley	Random	0.1042	0.0880	0.1206	0.0010	0.3969	1000
Amazon	ΔNDCG	Greedy seq. del.	Random	0.0140	0.0098	0.0184	0.0010	0.1976	1000
Amazon	ΔNDCG	LIME	Greedy seq. del.	0.0230	0.0175	0.0288	0.0010	0.2533	1000
Amazon	ΔNDCG	LIME	LOO	0.0018	0.0006	0.0033	0.0192	0.0818	1000
Amazon	ΔNDCG	LIME	Random	0.0370	0.0314	0.0429	0.0010	0.4000	1000
Amazon	ΔNDCG	LOO	Greedy seq. del.	0.0212	0.0158	0.0267	0.0010	0.2368	1000
Amazon	ΔNDCG	LOO	Random	0.0352	0.0296	0.0410	0.0010	0.3838	1000
Amazon	ΔNDCG	MC Shapley	Greedy seq. del.	0.0231	0.0177	0.0288	0.0010	0.2540	1000
Amazon	ΔNDCG	MC Shapley	LIME	0.0001	-0.0020	0.0023	0.8983	0.0039	1000
Amazon	ΔNDCG	MC Shapley	LOO	0.0019	-0.0006	0.0045	0.2598	0.0474	1000
Amazon	ΔNDCG	MC Shapley	Random	0.0371	0.0315	0.0429	0.0010	0.3982	1000
MovieLens	Success	Greedy seq. del.	Random	0.0286	0.0118	0.0460	0.0028	0.1025	1000
MovieLens	Success	LIME	Greedy seq. del.	0.1846	0.1606	0.2096	0.0010	0.4746	1000
MovieLens	Success	LIME	LOO	-0.0014	-0.0064	0.0040	0.6355	-0.0169	1000
MovieLens	Success	LIME	Random	0.2132	0.1898	0.2362	0.0010	0.5748	1000
MovieLens	Success	LOO	Greedy seq. del.	0.1860	0.1620	0.2100	0.0010	0.4747	1000
MovieLens	Success	LOO	Random	0.2146	0.1916	0.2378	0.0010	0.5705	1000
MovieLens	Success	MC Shapley	Greedy seq. del.	0.1712	0.1480	0.1950	0.0010	0.4550	1000
MovieLens	Success	MC Shapley	LIME	-0.0134	-0.0220	-0.0050	0.0066	-0.0974	1000
MovieLens	Success	MC Shapley	LOO	-0.0148	-0.0248	-0.0050	0.0066	-0.0923	1000
MovieLens	Success	MC Shapley	Random	0.1998	0.1774	0.2218	0.0010	0.5540	1000
MovieLens	ΔNDCG	Greedy seq. del.	Random	0.0068	0.0024	0.0113	0.0092	0.0959	1000
MovieLens	ΔNDCG	LIME	Greedy seq. del.	0.0367	0.0311	0.0425	0.0010	0.3887	1000

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Table S4 – continued from previous page

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
MovieLens	ΔNDCG	LIME	LOO	-0.0006	-0.0019	0.0006	0.3632	-0.0302	1000
MovieLens	ΔNDCG	LIME	Random	0.0435	0.0382	0.0491	0.0010	0.4964	1000
MovieLens	ΔNDCG	LOO	Greedy seq. del.	0.0373	0.0315	0.0433	0.0010	0.3904	1000
MovieLens	ΔNDCG	LOO	Random	0.0441	0.0386	0.0498	0.0010	0.4918	1000
MovieLens	ΔNDCG	MC Shapley	Greedy seq. del.	0.0344	0.0287	0.0401	0.0010	0.3742	1000
MovieLens	ΔNDCG	MC Shapley	LIME	-0.0023	-0.0040	-0.0007	0.0144	-0.0854	1000
MovieLens	ΔNDCG	MC Shapley	LOO	-0.0029	-0.0051	-0.0008	0.0144	-0.0847	1000
MovieLens	ΔNDCG	MC Shapley	Random	0.0413	0.0360	0.0467	0.0010	0.4840	1000

Table S5: conditional normalized-regret comparisons

Table S5. conditional normalized-regret comparisons.

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
Amazon	NRegret	Greedy seq. del.	Random	-0.4135	-0.6029	-0.2365	0.0010	-0.3143	196
Amazon	NRegret	LIME	Greedy seq. del.	-0.5862	-0.7598	-0.4379	0.0010	-0.5010	196
Amazon	NRegret	LIME	LOO	-0.0350	-0.0628	-0.0105	0.0267	-0.1853	196
Amazon	NRegret	LIME	Random	-0.9997	-1.1707	-0.8503	0.0010	-0.8633	196
Amazon	NRegret	LOO	Greedy seq. del.	-0.5512	-0.7268	-0.4014	0.0010	-0.4678	196
Amazon	NRegret	LOO	Random	-0.9647	-1.1438	-0.8120	0.0010	-0.8142	196
Amazon	NRegret	MC Shapley	Greedy seq. del.	-0.5776	-0.7595	-0.4286	0.0010	-0.4898	196
Amazon	NRegret	MC Shapley	LIME	0.0086	-0.0260	0.0431	0.6304	0.0348	196
Amazon	NRegret	MC Shapley	LOO	-0.0264	-0.0689	0.0154	0.4590	-0.0861	196
Amazon	NRegret	MC Shapley	Random	-0.9910	-1.1595	-0.8404	0.0010	-0.8664	196
MovieLens	NRegret	Greedy seq. del.	Random	0.0255	-0.2721	0.3421	0.8724	0.0089	339
MovieLens	NRegret	LIME	Greedy seq. del.	-1.2456	-1.6561	-0.8910	0.0010	-0.3453	339
MovieLens	NRegret	LIME	LOO	0.0081	-0.0112	0.0258	0.7967	0.0471	339
MovieLens	NRegret	LIME	Random	-1.2201	-1.4736	-0.9952	0.0010	-0.5432	339
MovieLens	NRegret	LOO	Greedy seq. del.	-1.2537	-1.6573	-0.8955	0.0010	-0.3461	339
MovieLens	NRegret	LOO	Random	-1.2281	-1.4810	-1.0027	0.0010	-0.5427	339
MovieLens	NRegret	MC Shapley	Greedy seq. del.	-1.2035	-1.6265	-0.8482	0.0010	-0.3337	339
MovieLens	NRegret	MC Shapley	LIME	0.0421	0.0173	0.0677	0.0064	0.1736	339
MovieLens	NRegret	MC Shapley	LOO	0.0501	0.0190	0.0818	0.0064	0.1694	339

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Table S5 – continued from previous page

Dataset	Metric	Left	Right	Δ	CI low	CI high	Holm-adjusted p	d_z	n
MovieLens	NRegret	MC Shapley	Random	-1.1780	-1.4312	-0.9610	0.0010	-0.5221	339

S4 Convergence matrix (Table S6)

Table S6. Supplementary convergence matrix. M_{pair} counts base permutations and T counts forward/reverse evaluated orders. Rank-valid fraction is the fraction of users with finite rank comparisons; threshold coverage is the fraction of those users satisfying both rank and action-overlap thresholds.

Dataset	Model	Utility	M_{pair}	T	Rank	Jaccard	Rank-valid	Coverage	Selected?
Amazon	itemknn	target margin	25	50	0.9348	0.8707	100.0%	43.0%	No
Amazon	itemknn	target margin	50	100	0.9532	0.9090	100.0%	60.0%	Yes
Amazon	itemknn	target margin	100	200	0.9690	0.9450	100.0%	70.0%	No
Amazon	itemknn	target margin	250	500	0.9739	0.9467	100.0%	78.0%	No
Amazon	itemknn	target margin	500	1000	0.9797	0.9617	100.0%	83.0%	No
Amazon	itemknn	target margin	1000	2000	0.9821	0.9653	100.0%	85.0%	No
Amazon	profile	target margin	25	50	0.9023	0.7913	100.0%	23.0%	No
Amazon	profile	target margin	50	100	0.9297	0.8563	100.0%	39.0%	No
Amazon	profile	target margin	100	200	0.9670	0.9110	100.0%	66.0%	Yes
Amazon	profile	target margin	250	500	0.9742	0.9297	100.0%	74.0%	No
Amazon	profile	target margin	500	1000	0.9841	0.9467	100.0%	78.0%	No
Amazon	profile	target margin	1000	2000	0.9888	0.9683	100.0%	89.0%	No
MovieLens	itemknn	ndcg	25	50	0.8257	0.6087	92.0%	22.8%	No
MovieLens	itemknn	ndcg	50	100	0.8575	0.6113	92.0%	27.2%	No
MovieLens	itemknn	ndcg	100	200	0.8847	0.6897	92.0%	39.1%	No
MovieLens	itemknn	ndcg	250	500	0.9167	0.7373	92.0%	52.2%	No
MovieLens	itemknn	ndcg	500	1000	0.9306	0.7740	92.0%	60.9%	No
MovieLens	itemknn	ndcg	1000	2000	0.9409	0.7813	92.0%	62.0%	No (max)
MovieLens	itemknn	target margin	25	50	0.9348	0.6467	100.0%	13.0%	No
MovieLens	itemknn	target margin	50	100	0.9584	0.7280	100.0%	33.0%	No
MovieLens	itemknn	target margin	100	200	0.9745	0.7880	100.0%	52.0%	No
MovieLens	itemknn	target margin	250	500	0.9853	0.8433	100.0%	69.0%	Yes
MovieLens	itemknn	target margin	500	1000	0.9902	0.8767	100.0%	74.0%	No
MovieLens	itemknn	target margin	1000	2000	0.9936	0.9040	100.0%	81.0%	No
MovieLens	profile	target margin	25	50	0.9455	0.7787	100.0%	25.0%	No
MovieLens	profile	target margin	50	100	0.9670	0.8320	100.0%	56.0%	Yes
MovieLens	profile	target margin	100	200	0.9779	0.8773	100.0%	70.0%	No
MovieLens	profile	target margin	250	500	0.9873	0.9053	100.0%	82.0%	No
MovieLens	profile	target margin	500	1000	0.9912	0.9267	100.0%	86.0%	No
MovieLens	profile	target margin	1000	2000	0.9941	0.9373	100.0%	89.0%	No

S5 Executable contract (Table S7)

Table S7. Expanded executable contract used by every final run.

Parameter	Frozen value
Temporal split	Last event test; penultimate event validation; ties sorted by timestamp and original record index.
Training histories	Complete histories are used to fit ItemKNN similarities and profile embeddings.
Scoring/player window	Most recent 20 training interactions; older training context is not added as fixed scoring context.
Candidates	Target plus 199 uniformly sampled unseen negatives; complete pre-test history including validation is excluded.
Models	ItemKNN with 200 top non-zero cosine neighbours; 64-dimensional profile aggregation robustness model.
Seeds	Model/attribution 42–46; candidate 1729; user 2718; tie 31415.
Attribution	Target-margin utility with top $L = 10$ competitors; $M_{\text{pair}} = 500$ and $T = 1000$ reverse-paired orders.
LIME	512 masks; normalized Hamming kernel width .25; ridge coefficient 1.0.
Intervention	Simultaneous $w_p \leftarrow 0.5w_p$; deletion and $\rho = .25$ sensitivity conditions.
Decision	No action, all singletons and all pairs; exact oracle at $B = 2$.
Inference	10,000 user bootstrap draws; 10,000 paired permutations; 1,000 matched AIA-null draws.

S6 Replication: sequential scorer, intervention-aware baselines, and prospective audits

To test the protocol under a nonlinear sequential scorer, selected audit components were replicated with a SASRec-style transformer (item and positional embeddings of dimension 64, one transformer-encoder layer with a single head and dropout 0.1, history truncated to the most recent 20 events, trained by leave-one-out BPR with Adam at learning rate 3×10^{-4} for 10 epochs; the user representation is the history-weighted mean of encoded positions, recomputed at scoring time so that masking/downweighting acts without retraining; Appendix S6). This is an architectural stress test, not evidence from a competitive recommender: neither the cohort recipe nor the longer full-corpus training variant exceeds the popularity reference under full-catalogue ranking (Table S20). The replication reports bounded-alignment, exact-estimator, interaction, prospective-audit, and action-selection diagnostics; it does not reproduce the complete primary method-by-metric matrix.

Three patterns emerge (Table S18). First, intervention-aware baselines—bounded binary/continuous LIME, finite differences, and integrated gradients, all trained or integrated along the same weight path used at deployment—reach bounded AIA 0.72–1.00 (bounded binary/continuous LIME 0.73–0.95, finite differences and integrated gradients 0.72–1.00), matching or exceeding binary-mask LIME (0.74–0.91) and Monte Carlo Shapley (0.14–0.70): part of binary LIME's weakness is therefore a perturbation-distribution mismatch, and Shapley's coalition averaging does not by itself predict bounded interventions better than path-matched baselines. Second, pair interactions $I_{pq} = \Delta(\{p, q\}) - \Delta(\{p\}) - \Delta(\{q\})$

are small relative to singleton effects for ItemKNN (mean ratio 0.07; additive-vs-realized pair ranking $\rho = 0.84$–0.90), empirically supporting the main manuscript's additive decision heuristic under ItemKNN, but are substantial under SASRec (ratios 0.21–0.78, additive-vs-realized $\rho = 0.45$–0.78), so interaction-aware selection is recommended whenever the scorer is strongly nonlinear. Third, replacing the held-out target by the model's own top-1 recommendation (a prospective, non-target-conditioned audit) leaves Shapley bounded alignment high (run means 0.73–0.87; 93–100% of users attain at least 0.5 alignment among the 126–227 users per run whose generated top-1 lies in the candidate set), showing that the attribution audit can also be instantiated without future-label conditioning. Finally, a component ablation on the same realized-effect currency shows that the protocol components matter: forcing an action (removing the no-action option) changes the selected action in 82–89% of users without improving the mean realized effect (mean Δ in $[-0.023, +0.001]$, with individual deltas reaching -0.51 under SASRec), and magnitude-only selection changes the action in 89–96% of users and degrades the mean realized effect under the sequential scorer (mean $\Delta = -0.013$ on Amazon, -0.041 on ML-1M), supporting the signed, abstention-aware action-selection rule in the main manuscript.

S6.1 Coalition and scorer-state budgets, and release provenance

Table S8 reports allocation-stage coalition/scorer-state budgets. With $m = |E_u|$ candidates and embedding dimension d, each scorer call costs $C_{\text{score}}(m, d)$, giving $O(T\, n_u\, C_{\text{score}})$ for Shapley, $O(S_{\text{LIME}}\, C_{\text{score}} + S_{\text{LIME}} n_u^2 + n_u^3)$ including the ridge solve, $O(n_u\, C_{\text{score}})$ for LOO, and $O(n_u^2\, C_{\text{score}})$ for greedy and the exact $B \leq 2$ oracle. On the replication machine the reverse-paired estimator costs 0.3–1.6 s per user (ItemKNN/SASRec) versus 0.03–0.12 s for bounded LIME, confirming that protocol overhead scales with scorer cost as analysed. Peak memory was not serialized and is therefore not claimed. These are analytical scorer-call counts, not claims that every call has identical wall-clock cost; the exact ItemKNN batch implementation reuses cached similarity blocks, while the joint-action oracle is evaluated once per user and shared across methods. Asymptotically in the player count n_u, the allocation stage costs $O(T\, n_u)$ cached prefix states for Monte Carlo Shapley with $T = 2M_{\text{pair}}$ evaluated orders, $O(S_{\text{LIME}})$ mask evaluations for LIME with $S_{\text{LIME}} = 512$, $O(n_u)$ states for leave-one-out, and $O(n_u^2)$ deletion evaluations for greedy sequential deletion; the shared exact $B \leq 2$ oracle adds $O(n_u^2)$ joint evaluations per user, computed once and reused by every method and utility. All budgets are therefore linear or quadratic in n_u and independent of catalogue size beyond the fixed candidate-set scoring block, which is cached per user.

Table S8. Attribution-stage coalition/scorer-state budget for a user with $n_u \leq 20$ players (multiply by $C_{\text{score}}(m, d)$ per call). Joint-action selection and the common exact oracle are excluded because they are shared outcome evaluations.

Method	Allocation-stage score budget	Reproducible definition
Monte Carlo Shapley	$T(n_u + 1) \leq 21{,}000$ prefix states before cache reuse	$M_{\text{pair}} = 500$ base orders plus reverses; coalition states cached and batched.
LIME	512 binary masks	Full, empty, all leave-one-out masks, then seeded random masks; weighted ridge with Hamming kernel.
LOO	$n_u + 1 \leq 21$ states	Full coalition and every one-deletion coalition; exact deletion diagnostic.
Greedy sequential deletion	$1 + n_u(n_u + 1)/2 \leq 211$ states	Every remaining deletion is recomputed at every step; lower-index ties win.
Random control	0 attribution utility calls	IID $\mathcal{N}(0, 1)$ scores; the common action/effect evaluation is still applied after selection.

The schema-v2 run summary records total wall-clock duration for the 80 recommendation runs: the 60 ItemKNN runs range from approximately 29.8 to 2,851.6 seconds (mean 183.0 seconds), and the 20 profile runs range from approximately 26.9 to 299.6 seconds (mean 95.0 seconds). These are complete-pipeline durations rather than isolated explainer timings; the analytical call-budget table above is the comparable per-method cost measure. The recorded environment is macOS-26.5.1-arm64, Python 3.12.13, NumPy 2.4.6, SciPy 1.17.1, pandas 2.3.3, scikit-learn 1.9.0, matplotlib 3.11.1, and PyYAML 6.0.3.

The submission package contains the editable LaTeX tables and vector figures used in the paper. The accompanying artifact will contain the complete generation code, configurations, preprocessing and execution scripts, validation reports, and machine-readable result matrices; its manifest uses repository-relative paths and content hashes. **The permanent or reviewer-view artifact URL must be inserted in the main manuscript and cover letter before submission.**

S7 Robustness Results

Table S9. Alignment robustness across the declared ItemKNN conditions: deletion and bounded AIA plus their difference for every method (absolute values shown for all).

Condition	Method	Del.	Bnd.	Gap	LIME bnd.	LOO bnd.	Rand. bnd.	Valid n
Amazon sampled	Shap.	0.286	0.414	+0.129	0.827	0.851	-0.010	993
Amazon sampled	LIME	0.930	0.827	-0.103	0.827	0.851	-0.010	993
Amazon sampled	LOO	1.000	0.851	-0.149	0.827	0.851	-0.010	993
Amazon sampled	Greedy	0.191	0.131	-0.060	0.827	0.851	-0.010	993
Amazon sampled	Rand.	-0.008	-0.010	-0.002	0.827	0.851	-0.010	993
Amazon full catalogue	Shap.	-0.288	-0.119	+0.169	0.722	0.790	-0.004	250
Amazon full catalogue	LIME	0.922	0.722	-0.200	0.722	0.790	-0.004	250
Amazon full catalogue	LOO	1.000	0.790	-0.210	0.722	0.790	-0.004	250
Amazon full catalogue	Greedy	0.573	0.449	-0.124	0.722	0.790	-0.004	250
Amazon full catalogue	Rand.	-0.001	-0.004	-0.003	0.722	0.790	-0.004	250
MovieLens sampled	Shap.	0.762	0.779	+0.017	0.933	0.978	0.000	1000
MovieLens sampled	LIME	0.951	0.933	-0.018	0.933	0.978	0.000	1000
MovieLens sampled	LOO	1.000	0.978	-0.022	0.933	0.978	0.000	1000
MovieLens sampled	Greedy	0.321	0.318	-0.002	0.933	0.978	0.000	1000
MovieLens sampled	Rand.	-0.001	0.000	+0.001	0.933	0.978	0.000	1000
MovieLens full catalogue	Shap.	0.574	0.618	+0.044	0.892	0.958	0.004	250
MovieLens full catalogue	LIME	0.926	0.892	-0.033	0.892	0.958	0.004	250
MovieLens full catalogue	LOO	1.000	0.958	-0.042	0.892	0.958	0.004	250
MovieLens full catalogue	Greedy	0.384	0.382	-0.003	0.892	0.958	0.004	250
MovieLens full catalogue	Rand.	0.003	0.004	+0.001	0.892	0.958	0.004	250
MovieLens 100 candidates	Shap.	0.809	0.821	+0.012	0.947	0.982	-0.003	250
MovieLens 100 candidates	LIME	0.960	0.947	-0.014	0.947	0.982	-0.003	250
MovieLens 100 candidates	LOO	1.000	0.982	-0.018	0.947	0.982	-0.003	250
MovieLens 100 candidates	Greedy	0.263	0.261	-0.003	0.947	0.982	-0.003	250
MovieLens 100 candidates	Rand.	-0.004	-0.003	+0.002	0.947	0.982	-0.003	250
MovieLens 500 candidates	Shap.	0.692	0.724	+0.032	0.922	0.971	0.004	250
MovieLens 500 candidates	LIME	0.944	0.922	-0.023	0.922	0.971	0.004	250
MovieLens 500 candidates	LOO	1.000	0.971	-0.029	0.922	0.971	0.004	250
MovieLens 500 candidates	Greedy	0.373	0.378	+0.005	0.922	0.971	0.004	250
MovieLens 500 candidates	Rand.	0.002	0.004	+0.002	0.922	0.971	0.004	250
MovieLens n_max=50	Shap.	0.800	0.808	+0.008	0.934	0.987	0.001	250
MovieLens n_max=50	LIME	0.945	0.934	-0.010	0.934	0.987	0.001	250
MovieLens n_max=50	LOO	1.000	0.987	-0.013	0.934	0.987	0.001	250
MovieLens n_max=50	Greedy	0.369	0.364	-0.006	0.934	0.987	0.001	250
MovieLens n_max=50	Rand.	0.001	0.001	+0.000	0.934	0.987	0.001	250
MovieLens n_max=100	Shap.	0.802	0.808	+0.006	0.918	0.987	0.001	250
MovieLens n_max=100	LIME	0.928	0.918	-0.010	0.918	0.987	0.001	250
MovieLens n_max=100	LOO	1.000	0.987	-0.013	0.918	0.987	0.001	250
MovieLens n_max=100	Greedy	0.387	0.381	-0.007	0.918	0.987	0.001	250
MovieLens n_max=100	Rand.	0.001	0.001	+0.000	0.918	0.987	0.001	250
MovieLens rho=0.25	Shap.	0.760	0.766	+0.007	0.950	0.992	-0.005	250
MovieLens rho=0.25	LIME	0.955	0.950	-0.005	0.950	0.992	-0.005	250
MovieLens rho=0.25	LOO	1.000	0.992	-0.008	0.950	0.992	-0.005	250
MovieLens rho=0.25	Greedy	0.378	0.377	-0.001	0.950	0.992	-0.005	250
MovieLens rho=0.25	Rand.	-0.004	-0.005	-0.001	0.950	0.992	-0.005	250

Table S10. Decision robustness on primary and representative boundary conditions. NRegret is conditional on active NDCG oracles; B=3 has no normalized-regret claim.

Condition	Method	ΔNDCG	Success	Abstention	NRegret	Active n
MovieLens primary	Shapley	.040	27.4%	0.0%	.268	339
MovieLens primary	LIME	.043	28.8%	0.0%	.225	339
MovieLens primary	LOO	.043	28.9%	0.0%	.217	339
Amazon primary	Shapley	.033	17.0%	1.0%	.206	196
Amazon primary	LIME	.033	17.1%	7.0%	.197	196
Amazon primary	LOO	.031	16.7%	8.6%	.232	196
MovieLens full catalogue	Shapley	.012	6.7%	–	.291	20
Amazon full catalogue	Shapley	.003	1.5%	–	.256	5
$B = 1$	Shapley	.026	–	–	–	–
$B = 1$	LIME	.026	–	–	–	–
$B = 1$	LOO	.027	–	–	–	–
$B = 3$	Shapley	.062	–	–	–	–
$B = 3$	LIME	.062	–	–	–	–
$B = 3$	LOO	.061	–	–	–	–

The archived B=3 sensitivity uses greedy rather than exhaustive triple enumeration and therefore reports no normalized regret. Full-catalogue rows are labelled exploratory: their active-oracle denominators ($n = 20$ MovieLens, $n = 5$ Amazon) are too small for inferential claims; an enlarged 1,000-user full-catalogue replication is provided in the release. "–" denotes a value intentionally omitted from this compact summary, not an unavailable result.

S8 Statistical Validation and Convergence

Table S11. Confirmatory paired comparisons. Bootstrap intervals are descriptive and unadjusted; significance is determined by Holm-adjusted permutation p-values.

Dataset	Metric	Comparison	Diff.	Unadj. CI	Holm p	d_z
Amazon	Gap	Shapley − LIME	.232	[.202,.263]	.0010	.473
Amazon	Gap	Shapley − LOO	.277	[.245,.311]	.0010	.525
Amazon	Bounded AIA	Shapley − LIME	-.413	[-.449,-.378]	.0010	-.726
Amazon	ΔNDCG	Shapley − LIME	.000	[-.002,.002]	.8983	.004
MovieLens	Gap	Shapley − LIME	.034	[.028,.043]	.0010	.289
MovieLens	Gap	Shapley − LOO	.039	[.032,.047]	.0010	.319
MovieLens	Bounded AIA	Shapley − LIME	-.155	[-.164,-.145]	.0010	-1.010
MovieLens	ΔNDCG	Shapley − LIME	-.002	[-.004,-.001]	.0144	-.085

Full pairwise comparisons will be included in the content-addressed machine-readable release.

Table S12. Selected convergence budgets and the unconverged NDCG stress test. Coverage means threshold coverage among rank-valid users.

Dataset	Model	Utility	M_{pair}	Rank	Jaccard	Rank-valid	Coverage
Amazon	ItemKNN	Margin	50	.953	.909	100%	60%
Amazon	Profile	Margin	100	.967	.911	100%	66%
MovieLens	ItemKNN	Margin	250	.985	.843	100%	69%
MovieLens	Profile	Margin	50	.967	.832	100%	56%
MovieLens	ItemKNN	NDCG	1000	.941	.781	92%	62%

The NDCG row is a declared stress test and is not used to select the primary target-margin budget.

The complete declared paired family, proportion CIs, quality-versus-popularity CIs, hierarchical user×seed sensitivity, and minimum-detectable-difference diagnostics are reproduced below from the release matrices (Tables S14–S17 below); Tables S3–S5 above cover the extended conditions.

Table S13. Primary signed diagnostic metrics for the ItemKNN target-margin analysis. This is a descriptive diagnostic summary; the confirmatory inferential targets are the paired comparisons in Table S14. Direction accuracy (main-manuscript Eq. 17) and top-3 precision use signed bounded effects, Precision@3 is the compound positive-and-correct-direction metric defined in main-manuscript Section 4.3, signed alignment is the Spearman correlation $\rho_S(-\phi^g_{u,p}, \Delta^{z,\mathrm{bnd}}_{u,p})$ between predicted benefits and bounded effects (seed-averaged per user, 10,000-draw bootstrap CIs), and stability is rank stability across the five seed runs. Direction accuracy and signed alignment are different quantities: the former is the fraction of non-zero effects whose sign is predicted correctly; the latter is a rank correlation over all players.

Dataset	Method	Dir. acc.	Prec.@3	Signed align.	Stab.	n
MovieLens	Shapley	.888	.820	0.929 [0.924,0.933]	.985	1000
MovieLens	LIME	.963	.920	0.974 [0.972,0.976]	.975	1000
MovieLens	LOO	.989	.967	0.994 [0.993,0.995]	1.000	1000
MovieLens	Greedy sequential deletion	.617	.296	0.342 [0.322,0.362]	1.000	1000
MovieLens	Random	.502	.165	0.004 [-0.003,0.011]	-.001	1000
Amazon	Shapley	.784	.603	0.855 [0.842,0.868]	.971	993
Amazon	LIME	.936	.596	0.949 [0.942,0.955]	.969	993
Amazon	LOO	.941	.596	0.966 [0.960,0.973]	1.000	993
Amazon	Greedy sequential deletion	.543	.361	0.109 [0.074,0.143]	1.000	993
Amazon	Random	.507	.277	0.009 [-0.005,0.024]	-.010	993

Table S14. Complete declared paired family for bounded AIA on the primary conditions: paired mean difference with user-bootstrap 95% CIs, two-sided sign-permutation p and paired d_z, Holm-corrected over the family.

Dataset	Model	Left	Right	n	Diff.	95% CI	Holm p	d_z
Amazon-Digital-Music	itemknn	Shapley	LIME	993	-0.413	[-0.449,-0.379]	0.0010	-0.726
Amazon-Digital-Music	itemknn	Shapley	LOO	993	-0.437	[-0.473,-0.403]	0.0010	-0.771
Amazon-Digital-Music	itemknn	Shapley	Greedy	993	0.283	[0.236,0.328]	0.0010	0.388
Amazon-Digital-Music	itemknn	Shapley	Random	993	0.424	[0.387,0.459]	0.0010	0.725
Amazon-Digital-Music	itemknn	LIME	LOO	993	-0.024	[-0.032,-0.016]	0.0010	-0.191
Amazon-Digital-Music	itemknn	LIME	Greedy	993	0.696	[0.665,0.729]	0.0010	1.317
Amazon-Digital-Music	itemknn	LIME	Random	993	0.837	[0.816,0.858]	0.0010	2.430
Amazon-Digital-Music	itemknn	LOO	Greedy	993	0.720	[0.689,0.753]	0.0010	1.380
Amazon-Digital-Music	itemknn	LOO	Random	993	0.861	[0.841,0.883]	0.0010	2.461
Amazon-Digital-Music	itemknn	Greedy	Random	993	0.141	[0.107,0.177]	0.0010	0.249
Amazon-Digital-Music	profile	Shapley	LIME	1000	-0.445	[-0.467,-0.424]	0.0010	-1.254
Amazon-Digital-Music	profile	Shapley	LOO	1000	-0.470	[-0.491,-0.450]	0.0010	-1.389
Amazon-Digital-Music	profile	Shapley	Greedy	1000	-0.155	[-0.183,-0.129]	0.0010	-0.350
Amazon-Digital-Music	profile	Shapley	Random	1000	0.306	[0.277,0.332]	0.0010	0.685
Amazon-Digital-Music	profile	LIME	LOO	1000	-0.025	[-0.028,-0.022]	0.0010	-0.493
Amazon-Digital-Music	profile	LIME	Greedy	1000	0.290	[0.277,0.303]	0.0010	1.336
Amazon-Digital-Music	profile	LIME	Random	1000	0.751	[0.733,0.768]	0.0010	2.667
Amazon-Digital-Music	profile	LOO	Greedy	1000	0.315	[0.301,0.329]	0.0010	1.392
Amazon-Digital-Music	profile	LOO	Random	1000	0.776	[0.758,0.793]	0.0010	2.679
Amazon-Digital-Music	profile	Greedy	Random	1000	0.461	[0.440,0.480]	0.0010	1.439
MovieLens-1M	itemknn	Shapley	LIME	1000	-0.155	[-0.165,-0.145]	0.0010	-1.010
MovieLens-1M	itemknn	Shapley	LOO	1000	-0.199	[-0.210,-0.189]	0.0010	-1.254
MovieLens-1M	itemknn	Shapley	Greedy	1000	0.460	[0.439,0.480]	0.0010	1.431
MovieLens-1M	itemknn	Shapley	Random	1000	0.778	[0.766,0.790]	0.0010	4.079
MovieLens-1M	itemknn	LIME	LOO	1000	-0.045	[-0.048,-0.042]	0.0010	-0.926
MovieLens-1M	itemknn	LIME	Greedy	1000	0.615	[0.597,0.633]	0.0010	2.183
MovieLens-1M	itemknn	LIME	Random	1000	0.933	[0.924,0.942]	0.0010	6.531
MovieLens-1M	itemknn	LOO	Greedy	1000	0.660	[0.642,0.677]	0.0010	2.388
MovieLens-1M	itemknn	LOO	Random	1000	0.978	[0.970,0.986]	0.0010	7.613
MovieLens-1M	itemknn	Greedy	Random	1000	0.318	[0.299,0.336]	0.0010	1.082
MovieLens-1M	profile	Shapley	LIME	1000	-0.186	[-0.191,-0.181]	0.0010	-2.284
MovieLens-1M	profile	Shapley	LOO	1000	-0.206	[-0.211,-0.201]	0.0010	-2.490
MovieLens-1M	profile	Shapley	Greedy	1000	0.471	[0.458,0.484]	0.0010	2.282
MovieLens-1M	profile	Shapley	Random	1000	0.787	[0.778,0.796]	0.0010	5.562
MovieLens-1M	profile	LIME	LOO	1000	-0.020	[-0.021,-0.019]	0.0010	-1.050
MovieLens-1M	profile	LIME	Greedy	1000	0.657	[0.645,0.669]	0.0010	3.335
MovieLens-1M	profile	LIME	Random	1000	0.973	[0.966,0.980]	0.0010	8.467
MovieLens-1M	profile	LOO	Greedy	1000	0.677	[0.665,0.689]	0.0010	3.515
MovieLens-1M	profile	LOO	Random	1000	0.993	[0.986,1.000]	0.0010	8.853
MovieLens-1M	profile	Greedy	Random	1000	0.316	[0.303,0.330]	0.0010	1.446

Table S15. Primary-cohort success and abstention proportions and paired quality difference against the popularity reference (Pop.), all on the 1,000-user primary ItemKNN cohort (993 on Amazon after constant-vector exclusion), seed-averaged within user with 10,000-draw user-bootstrap 95% CIs. Success is the fraction of users with positive seed-averaged realized NDCG effect; abstention is the fraction selecting no action. These replace the earlier replication-cohort (250-user) proportions so that all quality quantities are cohort-consistent with Table 3.

Dataset	Method	Quantity	Mean	95% CI	n
MovieLens-1M	Shapley	Success	0.274	[0.247,0.302]	1000
MovieLens-1M	Shapley	Abstention	0.000	[0.000,0.000]	1000
MovieLens-1M	Shapley	NDCG−Pop.	0.126	[0.106,0.145]	1000
MovieLens-1M	LIME	Success	0.288	[0.260,0.315]	1000
MovieLens-1M	LIME	Abstention	0.000	[0.000,0.000]	1000
MovieLens-1M	LIME	NDCG−Pop.	0.126	[0.106,0.145]	1000
MovieLens-1M	LOO	Success	0.289	[0.261,0.317]	1000
MovieLens-1M	LOO	Abstention	0.000	[0.000,0.000]	1000
MovieLens-1M	LOO	NDCG−Pop.	0.126	[0.106,0.145]	1000
MovieLens-1M	Greedy	Success	0.103	[0.085,0.122]	1000
MovieLens-1M	Greedy	Abstention	0.000	[0.000,0.000]	1000
MovieLens-1M	Greedy	NDCG−Pop.	0.126	[0.106,0.145]	1000
MovieLens-1M	Random	Success	0.074	[0.064,0.086]	1000
MovieLens-1M	Random	Abstention	0.000	[0.000,0.001]	1000
MovieLens-1M	Random	NDCG−Pop.	0.126	[0.106,0.145]	1000
Amazon-Digital-Music	Shapley	Success	0.170	[0.147,0.194]	1000
Amazon-Digital-Music	Shapley	Abstention	0.010	[0.004,0.017]	1000
Amazon-Digital-Music	Shapley	NDCG−Pop.	0.091	[0.066,0.115]	1000
Amazon-Digital-Music	LIME	Success	0.171	[0.148,0.195]	1000
Amazon-Digital-Music	LIME	Abstention	0.070	[0.055,0.086]	1000
Amazon-Digital-Music	LIME	NDCG−Pop.	0.091	[0.066,0.115]	1000
Amazon-Digital-Music	LOO	Success	0.167	[0.144,0.191]	1000
Amazon-Digital-Music	LOO	Abstention	0.086	[0.069,0.104]	1000
Amazon-Digital-Music	LOO	NDCG−Pop.	0.091	[0.066,0.115]	1000
Amazon-Digital-Music	Greedy	Success	0.090	[0.073,0.108]	1000
Amazon-Digital-Music	Greedy	Abstention	0.007	[0.002,0.012]	1000
Amazon-Digital-Music	Greedy	NDCG−Pop.	0.091	[0.066,0.115]	1000
Amazon-Digital-Music	Random	Success	0.066	[0.055,0.077]	1000
Amazon-Digital-Music	Random	Abstention	0.048	[0.041,0.055]	1000
Amazon-Digital-Music	Random	NDCG−Pop.	0.091	[0.066,0.115]	1000

Table S16. Paired sign-permutation tests on user-level success and abstention (primary cohort, seed-averaged indicators), Holm-corrected across the 12-test family. d is the mean paired difference.

Dataset	Left	Right	Quantity	Diff.	Holm p	n
MovieLens-1M	Shapley	LIME	Success	-0.0134	0.0216	1000
MovieLens-1M	Shapley	LIME	Abstention	+0.0000	1.0000	1000
MovieLens-1M	Shapley	LOO	Success	-0.0148	0.0216	1000
MovieLens-1M	Shapley	LOO	Abstention	+0.0000	1.0000	1000
MovieLens-1M	LIME	LOO	Success	-0.0014	1.0000	1000
MovieLens-1M	LIME	LOO	Abstention	+0.0000	1.0000	1000
Amazon-Digital-Music	Shapley	LIME	Success	-0.0010	1.0000	1000
Amazon-Digital-Music	Shapley	LIME	Abstention	-0.0600	0.0012	1000
Amazon-Digital-Music	Shapley	LOO	Success	+0.0032	1.0000	1000
Amazon-Digital-Music	Shapley	LOO	Abstention	-0.0758	0.0012	1000
Amazon-Digital-Music	LIME	LOO	Success	+0.0042	0.5039	1000
Amazon-Digital-Music	LIME	LOO	Abstention	-0.0158	0.0110	1000

Table S17. Sensitivity and power: hierarchical user×seed cluster-bootstrap CIs for primary bounded AIA means, and minimum detectable paired difference (MDE_{80}) for the Shapley–LIME contrast.

Dataset	Quantity	Value	CI low	CI high	*n*
Amazon-Digital-Music (itemknn)	Bounded AIA (Greedy, hier.)	0.131	0.100	0.162	993
Amazon-Digital-Music (itemknn)	Bounded AIA (LIME, hier.)	0.827	0.812	0.841	993
Amazon-Digital-Music (itemknn)	Bounded AIA (LOO, hier.)	0.851	0.834	0.866	993
Amazon-Digital-Music (itemknn)	Bounded AIA (Random, hier.)	-0.010	-0.024	0.004	993
Amazon-Digital-Music (itemknn)	Bounded AIA (Shapley, hier.)	0.414	0.381	0.447	993
Amazon-Digital-Music (profile)	Bounded AIA (Greedy, hier.)	0.465	0.452	0.478	1000
Amazon-Digital-Music (profile)	Bounded AIA (LIME, hier.)	0.755	0.747	0.764	1000
Amazon-Digital-Music (profile)	Bounded AIA (LOO, hier.)	0.780	0.771	0.789	1000
Amazon-Digital-Music (profile)	Bounded AIA (Random, hier.)	0.005	-0.013	0.019	1000
Amazon-Digital-Music (profile)	Bounded AIA (Shapley, hier.)	0.310	0.289	0.332	1000
MovieLens-1M (itemknn)	Bounded AIA (Greedy, hier.)	0.318	0.301	0.334	1000
MovieLens-1M (itemknn)	Bounded AIA (LIME, hier.)	0.933	0.928	0.938	1000
MovieLens-1M (itemknn)	Bounded AIA (LOO, hier.)	0.978	0.974	0.981	1000
MovieLens-1M (itemknn)	Bounded AIA (Random, hier.)	0.000	-0.006	0.007	1000
MovieLens-1M (itemknn)	Bounded AIA (Shapley, hier.)	0.779	0.769	0.789	1000
MovieLens-1M (profile)	Bounded AIA (Greedy, hier.)	0.317	0.304	0.329	1000
MovieLens-1M (profile)	Bounded AIA (LIME, hier.)	0.973	0.971	0.975	1000
MovieLens-1M (profile)	Bounded AIA (LOO, hier.)	0.994	0.992	0.995	1000
MovieLens-1M (profile)	Bounded AIA (Random, hier.)	0.000	-0.006	0.007	1000
MovieLens-1M (profile)	Bounded AIA (Shapley, hier.)	0.787	0.782	0.793	1000
Amazon-Digital-Music	$\mathrm{MDE}_8$0 (Shapley - LIME (bounded AIA))	0.032	–	–	1000
MovieLens-1M	$\mathrm{MDE}_8$0 (Shapley - LIME (bounded AIA))	0.008	–	–	1000

S9 Replication summary tables

Table S18. Replication aggregates per run: exact-Shapley validation of the MC estimator (mean over enumerated users of the per-user maximum absolute error, and mean rank ρ), interaction strength relative to singleton effects, additive-vs-realized pair ranking, and bounded/intervention-aware baseline alignment; last column is the prospective (non-target-conditioned) Shapley alignment. $--$ = undefined (constant exact spectrum or constant effect vectors).

Run	exact err	exact ρ	interact.	add./real.	Shap.	LIME bin.	prosp. AIA
Amazon ItemKNN	0.0001	0.783	0.078	0.842	0.669	0.738	0.727
Amazon SASRec	0.0129	0.395	0.782	0.453	0.140	0.942	0.798
ML-1M ItemKNN	0.0021	0.982	0.071	0.905	0.697	0.936	0.796
ML-1M SASRec	0.0184	0.688	0.206	0.785	0.241	0.936	0.870

Table S19. Extended analyses: unconditional (full-cohort) regret and prospective bounded AIA per explainer (non-target-conditioned audits), followed by point-by-point curves per run: equal-scorer-budget bounded AIA for Monte Carlo Shapley (budget in scored coalitions) and bounded LIME (budget in masks), the ρ-response of Monte Carlo Shapley, and the kernel-width (κ) sensitivity of bounded LIME.

Run	uncond. regret	prosp. Shap.	prosp. LIME b.	prosp. bLIME	prosp. FD	prosp. IG
Amazon ItemKNN	0.000	0.727	0.744	0.744	1.000	0.929
Amazon SASRec	0.017	0.798	0.824	0.912	0.610	0.542
ML-1M ItemKNN	0.001	0.796	0.933	0.947	0.974	0.969
ML-1M SASRec	0.003	0.870	0.848	0.937	0.920	0.906

Run	Shap. B=4.2k	Shap. B=10.5k	Shap. B=21k	Shap. B=42k	LIME 128	LIME 512	LIME 1024	LIME 2048
Amazon ItemKNN	0.708	0.708	0.708	0.714	0.754	0.757	0.758	0.759
Amazon SASRec	0.109	0.154	0.161	0.155	0.702	0.745	0.743	0.740
ML-1M ItemKNN	0.639	0.671	0.672	0.676	0.838	0.908	0.912	0.911
ML-1M SASRec	0.203	0.197	0.202	0.211	0.802	0.855	0.880	0.891

Run (ρ-response, Shapley)	$\rho = 0.0$	$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 0.9$
Amazon ItemKNN	0.708	0.708	0.708	0.708	0.708	0.708
Amazon SASRec	0.097	0.110	0.120	0.154	0.170	0.155
ML-1M ItemKNN	0.647	0.651	0.660	0.671	0.691	0.699
ML-1M SASRec	0.159	0.165	0.179	0.197	0.209	0.209

Run (κ-sensitivity, LIME)	$\kappa = 0.1$	$\kappa = 0.25$	$\kappa = 0.5$	$\kappa = 1.0$
Amazon ItemKNN	0.711	0.756	0.758	0.748
Amazon SASRec	0.893	0.936	0.877	0.822
ML-1M ItemKNN	0.957	0.936	0.961	0.958
ML-1M SASRec	0.963	0.939	0.946	0.938

Table S20. SASRec recommendation quality, five seeds; full-catalogue ranking against all unseen items on 1,000 users. Cohort rows use the replication training recipe (cohort-only histories, ten leave-one-out epochs); full-corpus rows train on every eligible user for thirty epochs with the chronological next-item objective. Neither configuration beats popularity, so the SASRec results in Table S18 demonstrate attribution transfer under a sequential scorer, not competitive recommendation; the quality gate's purpose is precisely to flag this, and the tuning gap is reported as a limitation.

Run	NDCG@10	HR@10	MRR	Pop. NDCG@10	Masking Δ	Gate
Amazon (cohort, 10 ep)	0.0002	0.0006	0.0009	0.0074	0.0004	0/5
Amazon (full corpus, 30 ep)	0.0002	0.0006	0.0009	0.0074	0.0004	1/5
ML-1M (cohort, 10 ep)	0.0019	0.0036	0.0030	0.0191	0.0029	5/5
ML-1M (full corpus, 30 ep)	0.0015	0.0040	0.0027	0.0191	0.0026	5/5

Table S21. Exact-Shapley validation of the reverse-paired estimator on enumerable users ($n_u \leq 12$): per-user maximum absolute error quantiles, top-2 signed-action Jaccard between exact- and MC-selected actions, and player-level sign-error rate.

Dataset	n	err p50	err p95	Jaccard mean	Jaccard p5	sign err.
Amazon	300	1.59e-04	6.71e-04	0.969	1.00	0.002
ML-1M	73	1.23e-03	2.36e-03	0.963	0.73	0.019

Table S22. LIME mask-design ablation (ItemKNN, 200 users, target margin): the seeded Bernoulli design with replacement, distinct masks without replacement, and full enumeration where feasible, plus the ridge penalty sweep under the Bernoulli design. Users whose attribution or effect vector is constant are excluded (valid/n).

Dataset	Design	Ridge α	valid/n users	Bounded AIA	SD
ML-1M	Bernoulli (primary)	1	400/400	0.906	0.092
ML-1M	Unique masks	1	200/200	0.907	0.092
ML-1M	Enumerate ($n_u \leq 9$)	1	7/7	0.930	0.093
ML-1M	Bernoulli	0.1	200/200	0.913	0.089
ML-1M	Bernoulli	10	200/200	0.794	0.169
Amazon	Bernoulli (primary)	1	144/400	0.669	0.205
Amazon	Unique masks	1	72/200	0.703	0.173
Amazon	Enumerate ($n_u \leq 9$)	1	47/164	0.776	0.124
Amazon	Bernoulli	0.1	72/200	0.663	0.244
Amazon	Bernoulli	10	72/200	0.673	0.176

Table S23. Profile-model variance components (bounded AIA, 100 users, $M_{\text{pair}} = 100$): fixed model across five attribution seeds versus five model-initialization seeds with fixed attribution.

Dataset	Attribution-seed var.	Model-seed var.	Cell mean
Amazon	0.0317	0.2333	0.283
ML-1M	0.0045	0.0232	0.675

Table S24. User-level convergence quantiles (100 users per dataset, independent $M_{\text{pair}} = 1000$ reference): median and 5th/25th percentiles of the per-user rank correlation to the reference, median budget-two action Jaccard, and the joint-threshold coverage (fraction of rank-valid users satisfying rank ≥ 0.95 and Jaccard ≥ 0.80).

Dataset	M_{pair}	rank med	rank p5	rank p25	Jaccard med	coverage
ML-1M ItemKNN	25	0.945	0.851	0.924	0.600	0.13
ML-1M ItemKNN	50	0.965	0.913	0.951	0.733	0.33
ML-1M ItemKNN	100	0.977	0.946	0.968	0.867	0.52
ML-1M ItemKNN	250	0.987	0.970	0.982	1.000	0.69
ML-1M ItemKNN	500	0.991	0.977	0.989	1.000	0.74
ML-1M ItemKNN	1000	0.995	0.989	0.992	1.000	0.81
Amazon ItemKNN	25	0.947	0.827	0.900	1.000	0.43
Amazon ItemKNN	50	0.968	0.840	0.937	1.000	0.60
Amazon ItemKNN	100	0.980	0.899	0.954	1.000	0.70
Amazon ItemKNN	250	0.991	0.896	0.965	1.000	0.78
Amazon ItemKNN	500	0.999	0.880	0.980	1.000	0.83
Amazon ItemKNN	1000	1.000	0.899	0.985	1.000	0.85

Table S25. KernelSHAP baseline (Shapley-kernel weighted least squares with the efficiency constraint) against the reverse-paired Monte Carlo Shapley estimator, bounded AIA, primary ItemKNN conditions, $\rho = 0.5$, user-level paired comparison with 10,000-draw bootstrap CIs. KernelSHAP uses 256/512 masks, the same mask budget as the LIME comparison; Monte Carlo Shapley uses $M_{\text{pair}} = 250$ ($T = 500$ evaluated orders). Both estimators agree qualitatively: Shapley bounded alignment remains far below LIME (0.93/0.83) and LOO (0.98/0.85) under either estimator.

Dataset	Masks	n	MC Shap.	KernelSHAP	Diff.	d_z
MovieLens-1M	256	1000	0.744	0.650	+0.094 [+0.086,+0.102]	+0.72
MovieLens-1M	512	1000	0.744	0.664	+0.080 [+0.073,+0.088]	+0.66
Amazon	256	810	0.607	0.635	-0.029 [-0.035,-0.023]	-0.32
Amazon	512	810	0.607	0.633	-0.026 [-0.032,-0.020]	-0.31

Table S26. LightGCN recommendation-quality gate (full-corpus training, tuned: deeper propagation, larger embeddings, higher learning rate, 80 epochs, five seeds). LightGCN is adapted to the protocol's inference-time history-weighting interface (user vector recomputed as the weighted mean of propagated item embeddings). Even after tuning it does not beat the popularity reference on NDCG@10 on either dataset, so—like the SASRec-style replication—it serves as an architecture-applicability cell rather than a competitive-recommender robustness cell.

Dataset	NDCG@10	Popularity	Gate passes
MovieLens-1M	0.0042	0.0191	5/5 masking
Amazon	0.0035	0.0074	4/5 masking

Table S27. Generalization of the central finding to a graph architecture (LightGCN) and a third dataset (Gowalla location check-ins): bounded AIA per attribution method. Path-matched baselines (finite differences, integrated gradients, bounded LIME) exceed Monte Carlo Shapley on every configuration, replicating the primary ItemKNN pattern.

Configuration	Shapley	LIME (binary)	LIME (bnd.)	Fin. diff.	Int. grad.
MovieLens-1M, LightGCN	0.733	0.887	0.949	0.950	0.942
Amazon, LightGCN	0.234	0.806	0.888	0.870	0.853
Gowalla, ItemKNN	0.663	0.734	0.722	0.992	0.977

Table S28. Monte Carlo error control at the $n_{\max} = 20$ cap: per-user Monte Carlo standard errors of the Shapley values relative to the largest $|\phi|$, and rank correlation between the selected convergence-budget estimate ($M_{\text{pair}} = 250$) and an independent larger-budget reference ($M_{\text{pair}} = 1000$, separate seed stream).

Dataset	n users	Rel. SE mean	Rel. SE med.	Cross-budget corr. mean	min
MovieLens-1M	50	0.079	0.073	0.989	0.948
Amazon	50	0.062	0.059	0.997	0.982

Table S29. Exhaustive budget-B oracle (all $\leq B$ actions enumerated, replacing the greedy lower bound): oracle effect and additive-selection regret, $B = 2$ and $B = 3$ (1,351 actions at $n = 20$). Additive selection attains low regret at both budgets.

Dataset	B	n users	Oracle effect	Additive regret
MovieLens-1M	2	200	0.0026	0.0001
MovieLens-1M	3	200	0.0035	0.0011
Amazon	2	200	0.0020	0.0000
Amazon	3	200	0.0023	0.0003

Extended empirical basis

Five additional experiment families extend the empirical basis beyond the primary ItemKNN analysis (Tables S26–S29).

Architecture and dataset generality. The central finding—that path-matched baselines (finite differences, integrated gradients, bounded LIME) exceed Monte Carlo Shapley on bounded-intervention alignment—replicates under a graph architecture (LightGCN, Table S27) and on a third dataset from a different domain (Gowalla location check-ins). LightGCN is adapted to the protocol's inference-time history-weighting interface; it does not pass the recommendation-quality gate (Table S26), so it is reported as an architecture-applicability cell rather than a competitive-recommender cell. A tuning sweep (deeper propagation, larger embeddings, higher learning rate, 80 epochs, full-corpus training) leaves NDCG@10 below the popularity reference on both datasets, suggesting that the scoring-time user-vector interface required for bounded downweighting, rather than the tested training budget, remains a limiting factor. Obtaining a competitive neural recommender that also exposes the bounded history-weighting interface therefore remains an open engineering requirement.

Monte Carlo error control at the cap. The earlier exact-validation gap— convergence was only validated exactly for $n_u \le 12$ while 83.1% of MovieLens users sit at the $n_{\max} = 20$ cap—is addressed by per-user Monte Carlo standard errors plus an independent larger-budget reference (Table S28): relative standard errors are small (mean 0.06–0.08 of the largest $|\phi|$) and the selected convergence-budget estimate agrees with the independent $M_{\text{pair}} = 1000$ reference at rank correlation ≥ 0.95 for every user (mean ≥ 0.989).

Exhaustive budget-3 oracle. The greedy budget-3 lower bound is replaced by exhaustive enumeration of all ≤ 3 actions (1,351 at $n = 20$, Table S29); additive selection attains low regret at both $B = 2$ and $B = 3$.

Worked example. A representative MovieLens user (target item 1344, $n_u = 20$) illustrates the knowledge-level output: the two most-beneficial interactions by attribution (items 336 and 2094, both negative ϕ) are selected by the additive rule, and the resulting joint action coincides exactly with the exact budget-2 oracle (realized target-margin effect +0.0031).

S10 Computational Cost and Reproducibility

Table S30. Computational cost and release record. Analytical budgets are per user; runtime values are complete-pipeline measurements.

Item	Value
Shapley budget	$T(n_u + 1) \le 21{,}000$ prefix states before cache reuse.
LIME budget	512 masks.
LOO budget	$n_u + 1 \le 21$ states.
Greedy budget	$1 + n_u(n_u + 1)/2 \le 211$ states.
Exact oracle budget	$1 + n_u + \binom{n_u}{2} = 211$ realized-effect evaluations at $B = 2$ (1,351 at $B = 3$), independent of the explainer.
ItemKNN runtime	29.8–2851.6 s; mean 183.0 s over 60 runs.
Profile runtime	26.9–299.6 s; mean 95.0 s over 20 runs.
Environment	macOS-26.5.1-arm64; Python 3.12.13; NumPy 2.4.6; SciPy 1.17.1; pandas 2.3.3; scikit-learn 1.9.0; matplotlib 3.11.1; PyYAML 6.0.3.
Run inventory	80 recommendation runs; 5 convergence studies; the accompanying artifact will include a content-addressed manifest.

Complete per-user, per-seed, robustness, convergence, and pairwise-comparison matrices will be provided in the content-addressed CSV/JSON release. The historical schema did not record processor model, RAM, or peak RSS; the current runner records those fields for future regenerated releases.

Complexity analysis

Let $n = n_u$ players, $m = |E_u|$ candidates, K the neighbour cap (200), $\bar{K} \le K$ the mean retained neighbours per candidate, d the embedding dimension, and $T = 2M_{\text{pair}}$ evaluated orders. One scorer call costs $O(nK + m\bar{K})$ for ItemKNN (weighted neighbour aggregation plus candidate scoring) and $O((n+m)d)$ for the profile model; a bounded intervention changes weights only, and the protocol re-scores from scratch per action so audit cost equals one scorer call. Attribution then costs, in scorer calls per user: Monte Carlo Shapley $O(T(n+1))$ (prefix walks with a coalition cache shared between each order and its reverse), LIME $O(m_{\text{mask}})$, leave-one-out $O(n+1)$, greedy $O(n^2)$, random $O(1)$; the exact $B = 2$ oracle adds $1 + n + \binom{n}{2}$ realized-effect evaluations, and each utility evaluation is $O(m)$ after scoring. With $n \le 20$, $T \le 1000$, $m = 201$ (primary) the Shapley audit dominates ($\le 21{,}000$ calls); under full-catalogue evaluation m grows to 10^3–10^4 and the per-call candidate term dominates linearly. Memory is $O(m+n)$ per user for scores, effects, and attributions; the Shapley cache stores scalar coalition utilities for visited prefix states, $O(T(n+1))$ entries. Measured per-user median wall time on MovieLens-1M ItemKNN (replication runner, three repeats): Shapley 0.65 s, LIME 0.043 s, LOO 0.0018 s; these are method-specific attribution times, complementary to the complete-pipeline runtimes in the table above.